

FERNS AND PETALS (FNP) SALES ANALYSIS

Executive Summary and Business Intelligence Report

Prepared as a Data Analytics Portfolio Project

Tool Used: Microsoft Excel

Analysis Focus: Sales Performance, Customer Behavior, Product Performance, and Business Insights

1. EXECUTIVE SUMMARY

This project presents a comprehensive analysis of the **Ferns and Petals (FNP) sales dataset**, with the objective of transforming raw transactional data into meaningful business insights that can support strategic decision-making.

Ferns and Petals is a gifting business that specializes in products purchased for a variety of occasions, including **Diwali, Raksha Bandhan, Holi, Valentine's Day, Birthdays, and Anniversaries**. The nature of the gifting industry makes sales performance highly dependent on customer preferences, seasonal demand, occasions, product popularity, geographic location, and timely delivery.

Using **Microsoft Excel**, the dataset was analyzed, cleaned, transformed, and visualized to identify patterns in sales performance and customer purchasing behavior. An interactive Excel dashboard was developed to provide a consolidated view of key business performance indicators and enable stakeholders to explore the data from different perspectives.

The analysis focuses on ten major business questions covering **overall revenue, order and delivery performance, monthly sales trends, product revenue contribution, customer spending, top-performing products, geographic order distribution, order quantity and delivery time, occasion-based revenue, and product popularity by occasion**.

The resulting dashboard provides a practical decision-support tool that allows business stakeholders to move beyond basic sales reporting and gain a deeper understanding of the factors driving business performance.

The analysis demonstrates how Excel can be effectively used as a business intelligence and data analytics tool to support **sales strategy, inventory planning, customer segmentation, marketing decisions, and operational improvement.**

2. PROJECT OBJECTIVE

The primary objective of this project was to analyze FNP's sales data and identify actionable insights that can help the company improve its sales strategy, optimize product offerings, understand customer behavior, and enhance customer satisfaction.

The analysis was designed to answer the following key business questions:

1. What is the overall revenue generated by the business?
2. What is the average order and delivery time?
3. How does sales performance fluctuate across the months of 2023?
4. Which products generate the highest revenue?
5. How much are customers spending on average?
6. How are the top five products performing in terms of sales?
7. Which ten cities have the highest number of orders?
8. Does a higher order quantity affect delivery time?
9. How does revenue compare across different occasions?
10. Which products are most popular for specific occasions?

Answering these questions provides management with a holistic view of both **commercial performance and operational efficiency.**

3. DATA ANALYSIS APPROACH

The analysis was conducted entirely using **Microsoft Excel.**

The project involved several stages of data analysis, beginning with an examination of the available transactional data and progressing through data preparation, analysis, visualization, and dashboard development.

The general analytical workflow included:

Data Preparation

The raw dataset was reviewed and prepared for analysis to ensure that the relevant fields were available and suitable for calculations and visualization.

Data Cleaning

The data was checked for inconsistencies, missing values, formatting issues, and other potential data-quality problems that could affect the accuracy of the analysis.

Data Transformation

Relevant date fields and business measures were structured to support analysis by month, occasion, product, city, and other relevant dimensions.

Data Analysis

Excel functions, calculations, PivotTables, and analytical techniques were used to calculate key performance indicators and identify trends and patterns.

Data Visualization

Charts and visual summaries were created to communicate the results clearly and make important business insights easier to interpret.

Dashboard Development

The individual analyses were consolidated into an interactive Excel dashboard that provides a centralized view of FNP's sales performance.

4. KEY PERFORMANCE INDICATORS

The dashboard focuses on key metrics that provide management with an immediate overview of business performance.

The major KPIs include:

- **Total Revenue**
- **Total Orders**
- **Average Order Value**
- **Average Delivery Time**
- **Average Customer Spending**
- **Total Quantity Sold**

These KPIs provide a high-level snapshot of the business while allowing users to explore the underlying factors contributing to the results.

The KPI approach is particularly useful for management because it enables decision-makers to quickly identify the current performance level before examining detailed trends and patterns.

5. OVERALL REVENUE ANALYSIS

The first major business question addressed by the project is the company's **overall revenue performance**.

Total revenue provides the broadest measure of financial performance within the dataset and serves as a foundation for understanding the contribution of individual products, occasions, customers, cities, and time periods.

However, total revenue alone does not explain why the business is performing at a particular level. For this reason, the analysis breaks revenue down into several dimensions.

The dashboard allows management to investigate revenue according to:

- Product
- Month
- Occasion
- City

- Customer purchasing behavior

This approach makes it possible to identify the major drivers of revenue and determine where the company should focus its resources.

The revenue analysis should therefore be interpreted together with product and occasion performance. A product that generates significant revenue may require additional inventory investment, while a high-performing occasion may require increased marketing and fulfillment capacity.

6. AVERAGE ORDER AND DELIVERY TIME

The analysis of order and delivery time is particularly important for a gifting company because customers often purchase products for specific events and dates.

For many customers, the value of a gift depends not only on the quality of the product but also on whether it arrives at the expected time.

The analysis therefore evaluates the time between order placement and delivery to understand the efficiency of the fulfillment process.

Monitoring delivery time can help FNP identify:

- Potential delivery delays.
- Operational bottlenecks.
- Locations with longer delivery times.
- Periods of increased operational pressure.
- The potential impact of large orders on fulfillment.

The findings can be used to improve delivery planning and ensure that the company maintains reliable service, especially during high-demand occasions.

Improving delivery reliability can contribute directly to customer satisfaction and increase the likelihood of repeat purchases.

7. MONTHLY SALES PERFORMANCE IN 2023

The monthly sales analysis examines how revenue and sales activity changed throughout 2023.

The gifting industry is highly seasonal, and customer purchasing behavior is often influenced by festivals, holidays, birthdays, anniversaries, and other special occasions.

Analyzing sales by month allows FNP to identify:

- High-performing months.
- Low-performing months.
- Seasonal fluctuations.
- Potential demand peaks.
- Opportunities for targeted marketing campaigns.

This information is particularly valuable for planning inventory and logistics.

During months with historically high demand, FNP can prepare by increasing product availability, strengthening delivery capacity, and launching marketing campaigns in advance.

During lower-demand periods, the company can use promotional campaigns, discounts, product bundles, and targeted customer engagement strategies to stimulate sales.

The monthly analysis therefore provides a foundation for **seasonal sales forecasting and resource planning**.

8. TOP PRODUCTS BY REVENUE

Product performance is a critical component of the analysis because different products contribute differently to total business revenue.

The **Top Products by Revenue** analysis identifies the products that generate the greatest financial contribution.

These products should receive particular attention from management because poor availability or inadequate promotion of high-performing products could negatively affect overall revenue.

The results can support decisions related to:

- Inventory management.
- Product promotion.
- Product bundling.
- Pricing strategy.
- Marketing investment.

However, revenue should not be considered the only measure of product success.

A complete product-performance evaluation should ideally consider revenue alongside:

- Sales volume.
- Profit margin.
- Order frequency.
- Customer satisfaction.
- Repeat purchase behavior.

This broader perspective helps FNP identify products that are not only popular but also strategically valuable to the business.

9. CUSTOMER SPENDING ANALYSIS

Understanding customer spending behavior is essential for developing effective customer segmentation and retention strategies.

The analysis examines how much customers spend and provides insight into the average value generated by individual customers and transactions.

Customer spending analysis can help the company distinguish between:

- High-value customers.

- Medium-value customers.
- Low-value customers.

High-value customers represent an important opportunity for customer retention and loyalty initiatives.

FNP could consider offering these customers:

- Personalized product recommendations.
- Loyalty rewards.
- Exclusive promotions.
- Premium gift collections.
- Early access to seasonal offers.

For customers with lower average spending, the company can focus on increasing basket size through:

- Product bundles.
- Cross-selling.
- Complementary product recommendations.
- Quantity-based promotions.

The ultimate objective is to increase **customer lifetime value** while maintaining a positive customer experience.

10. SALES PERFORMANCE OF THE TOP 5 PRODUCTS

The analysis of the **Top 5 Products** provides a focused view of the products that contribute most strongly to sales performance.

Monitoring the top five products over time allows management to determine whether their performance is consistent or whether demand varies depending on the month or occasion.

If a small number of products account for a large proportion of total sales, FNP should ensure that these products remain consistently available.

At the same time, the company should avoid becoming excessively dependent on a limited number of products.

A balanced strategy would involve:

- Protecting the performance of existing best sellers.
- Maintaining sufficient inventory.
- Identifying emerging products with growth potential.
- Promoting underperforming but profitable products.
- Diversifying the product portfolio.

This approach can help the company maintain stable revenue while reducing the risks associated with product concentration.

11. TOP 10 CITIES BY NUMBER OF ORDERS

The geographic analysis identifies the **Top 10 Cities by Number of Orders**.

This analysis provides insight into where FNP's customer demand is concentrated.

Understanding geographic demand is important for both commercial and operational planning.

Cities with high order volumes may justify:

- Increased marketing investment.
- Improved delivery infrastructure.
- Higher inventory availability.
- Faster fulfillment options.
- Location-specific promotions.

The analysis can also support market expansion decisions.

Cities with growing demand but relatively limited operational coverage may represent potential opportunities for future expansion.

By combining geographic demand with delivery performance, FNP can identify areas where improvements in logistics could have the greatest impact on customer satisfaction.

12. ORDER QUANTITY VS. DELIVERY TIME

One of the operational questions addressed in the analysis is whether the quantity of products included in an order affects delivery time.

The relationship between order quantity and delivery time can provide insight into the scalability of FNP's fulfillment process.

If larger orders consistently require more time to process and deliver, this may indicate operational constraints involving:

- Inventory availability.
- Order preparation.
- Packaging.
- Processing capacity.
- Transportation.
- Last-mile delivery.

If such a relationship is identified, FNP could consider introducing dedicated processes for larger orders.

Potential improvements include:

- Advance scheduling for bulk orders.
- Dedicated bulk-order processing.

- Inventory reservation.
- Additional staff during peak periods.
- Improved order prioritization.

This is especially relevant for large corporate orders, events, and high-volume gifting requirements.

13. REVENUE COMPARISON BY OCCASION

Occasion-based revenue analysis is particularly important for FNP because customer purchases are strongly influenced by the reason for giving the gift.

The analysis compares revenue generated across different occasions, including events such as:

- Diwali.
- Raksha Bandhan.
- Holi.
- Valentine's Day.
- Birthdays.
- Anniversaries.

This analysis helps identify which occasions represent the strongest commercial opportunities.

The results can guide:

- Marketing budget allocation.
- Promotional planning.
- Inventory preparation.
- Product development.
- Seasonal campaign scheduling.

High-performing occasions should receive sufficient inventory and operational support to avoid stock shortages and delivery delays.

Lower-performing occasions may require targeted marketing strategies to increase customer awareness and stimulate demand.

It is also important to distinguish between **order volume and revenue value**. An occasion may generate many orders but relatively low revenue, while another occasion may produce fewer orders with a significantly higher average order value.

14. PRODUCT POPULARITY BY OCCASION

The final major analysis examines the relationship between products and occasions.

This analysis determines which products are most popular for specific gifting occasions.

The results provide FNP with an opportunity to develop a more personalized and targeted sales strategy.

Rather than promoting the same products to every customer, the company can create occasion-specific product recommendations.

Potential strategies include:

- Occasion-specific product collections.
- Personalized product recommendations.
- Targeted advertising campaigns.
- Occasion-based gift bundles.
- Customized online shopping experiences.
- Cross-selling recommendations.

Understanding product preferences by occasion can also improve inventory planning.

For example, products that consistently perform well during a specific festival should be stocked adequately before the relevant season begins.

This reduces the risk of stockouts while ensuring that customers can find the products they are most likely to purchase.

15. KEY BUSINESS INSIGHTS

The analysis demonstrates that FNP's business performance is influenced by multiple interconnected factors.

The most important insights from the project include:

Revenue is influenced by multiple dimensions

Overall revenue should be analyzed alongside product, occasion, location, and time-based performance to identify the actual drivers of business growth.

Sales are likely to be seasonal

Monthly and occasion-based analysis is essential for understanding fluctuations in demand and preparing for peak periods.

Product performance varies

A limited number of products may contribute disproportionately to revenue, making inventory planning and product diversification important.

Customer spending behavior provides segmentation opportunities

Customers with different spending patterns require different engagement and retention strategies.

Geographic demand is concentrated

The Top 10 Cities analysis provides opportunities for targeted marketing and logistics optimization.

Delivery performance affects customer experience

Reliable and timely delivery is critical in the gifting industry, where purchases are often associated with specific dates.

Occasion-based personalization can improve sales

Understanding which products customers prefer for specific occasions can help FNP deliver more relevant recommendations and promotions.

16. BUSINESS RECOMMENDATIONS

Based on the analytical framework developed in Excel, the following recommendations are proposed for FNP.

16.1 Improve Seasonal Demand Planning

Use historical monthly and occasion-based sales patterns to forecast demand and prepare inventory, staffing, marketing, and delivery capacity ahead of peak periods.

16.2 Prioritize High-Revenue Products

Maintain sufficient inventory for top-performing products and ensure that these products receive appropriate promotional visibility.

16.3 Develop Occasion-Specific Marketing

Create targeted campaigns based on customer preferences associated with individual occasions rather than applying a single marketing strategy across all customer segments.

16.4 Improve Delivery Efficiency

Monitor delivery time by city, order quantity, and occasion to identify potential bottlenecks and improve fulfillment performance.

16.5 Increase Average Customer Spending

Use bundles, cross-selling, complementary products, and personalized recommendations to encourage customers to increase their average order value.

16.6 Strengthen Customer Retention

Identify high-value customers and introduce loyalty programs and personalized engagement strategies designed to encourage repeat purchases.

16.7 Optimize Geographic Expansion

Focus marketing and operational investment on high-order cities while evaluating emerging cities for potential expansion.

16.8 Diversify the Product Portfolio

Continue supporting high-performing products while developing and promoting additional products to reduce dependency on a small number of best sellers.

16.9 Align Inventory With Occasions

Use historical product popularity by occasion to ensure that the right products are available at the right time.

16.10 Adopt Continuous Data Monitoring

The Excel dashboard should be updated regularly as new sales data becomes available. Continuous monitoring will enable FNP to detect changes in customer behavior, sales patterns, and operational performance.

17. CONCLUSION

The **Ferns and Petals Sales Analysis Project** demonstrates the value of using data analytics to convert transactional sales data into actionable business intelligence.

Through the use of **Microsoft Excel**, the project provides a structured analysis of FNP's sales performance across revenue, products, customers, cities, months, occasions, and delivery operations.

The dashboard enables stakeholders to understand not only **how much the company is selling**, but also **what customers are buying, when they are buying, where demand is concentrated, which occasions drive revenue, and how efficiently orders are fulfilled**.

The analysis provides a strong foundation for improving sales planning, inventory management, marketing effectiveness, customer segmentation, and delivery operations.

The most important strategic opportunity for FNP is to adopt a more data-driven approach to decision-making. By combining historical sales trends with customer spending patterns, product

performance, geographic demand, and occasion-based preferences, the company can better anticipate customer needs and allocate resources more effectively.

Ultimately, the insights generated through this project can help FNP pursue a balanced strategy focused on **revenue growth, operational efficiency, customer satisfaction, and long-term customer loyalty.**

The Excel dashboard serves as a practical business intelligence solution that allows management to monitor performance, identify opportunities, and make informed decisions based on evidence rather than assumptions.

18. PROJECT TECHNOLOGY

Primary Tool: Microsoft Excel

Excel Features and Techniques Applied:

- Data Cleaning
- Data Preparation
- Data Transformation
- Excel Formulas and Functions
- PivotTables
- PivotCharts
- Interactive Dashboard Design
- KPI Development
- Data Visualization
- Trend Analysis
- Comparative Analysis
- Business Intelligence Reporting

19. PROJECT DELIVERABLE

The final project consists of:

1. Raw Sales Dataset

The original FNP transactional sales data used as the foundation for the analysis.

2. Cleaned and Prepared Data

The processed dataset prepared for analytical calculations and visualization.

3. Excel Analysis

The calculations, PivotTables, and analytical components used to answer the business questions.

4. Interactive Excel Dashboard

A consolidated dashboard presenting key performance indicators and visual insights.

5. Executive Summary

This report documenting the analytical approach, key business questions, insights, and strategic recommendations.

20. PORTFOLIO PROJECT SUMMARY

Project Title: Ferns and Petals (FNP) Sales Analysis

Role: Data Analyst

Tool: Microsoft Excel

Domain: Sales and Business Analytics

Objective: Analyze sales data to identify revenue trends, customer behavior, product performance, geographic demand, delivery efficiency, and occasion-based purchasing patterns.

Key Skills Demonstrated:

- Data Analysis
- Data Cleaning

- Excel Analytics
- PivotTables
- Data Visualization
- Dashboard Development
- KPI Analysis
- Sales Performance Analysis
- Customer Behavior Analysis
- Business Insight Generation
- Strategic Recommendation

Business**Value:**

The project demonstrates the ability to transform raw transactional data into a structured analytical dashboard and executive-level insights that can support data-driven decisions in sales strategy, marketing, inventory management, customer retention, and operational planning.